

Minura Ashen Samaramanna

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PERSONAL STATEMENT

I am an Electronic and Telecommunication Engineering undergraduate passionate about mathematics and emerging technologies, with a mission to acquire knowledge, approach problems creatively, and develop solutions that are both practical and efficient, grounded in rigorous mathematical reasoning and analysis.

EDUCATION

University of Moratuwa *Feb 2023 - Present*
B.Sc.(Hons) in Electronic and Telecommunication Engineering Moratuwa, Sri Lanka

- CGPA: 3.93/4.0
- Dean's List for Semester 1,2,4,5

Balangoda Ananda Maithreya National School *Feb 2013 - Jan 2022*
G.C.E. Advanced Level (Physical Science Stream) Ratnapura, Sri Lanka

- 4A's (Combined Mathematics, Physics, Chemistry, General English)
- District Rank: 1 | Island Rank: 28 | Z-Score: 2.7031

EXPERIENCE

Software Engineer Intern – WSO2 LLC *Nov 2025 – May 2026*
Integration AI Team Colombo, Sri Lanka

Semantic Code Search Tool for MI Copilot [Search Tool](#) 

- Evaluated lightweight embedding models on WSO2 Micro Integrator XML configurations.
- Designed and implemented a background embedding service to continuously index configuration changes.
- Built a semantic retrieval pipeline to enable intent-aware code search, overcoming expensive brute-force context retrieval for natural language queries.

Tech Stack: TypeScript, Node.js, Orama DB

Exporting Salesforce Leads to Google Sheets Automation [Prebuilt Integration](#) 

- Developed an automated integration using Ballerina to periodically export filtered Salesforce leads to Google Sheets via cron job scheduling.
- Implemented dynamic user configurations, enabling custom lead filtering and selectable synchronization modes (append or full replace).

Tech Stack: Ballerina, Salesforce API, Google Sheets API

PROJECTS

Strength Training Exercise Recognition System [Fitness-Tracker](#) 

- Developed a context-aware strength training tracker using wristband accelerometer and gyroscope data
- Applied supervised learning algorithms to classify exercises, count repetitions, and detect improper form
- Collected and processed multi-participant sensor dataset for model training and evaluation sessions.

Hybrid Geometric-Generative Navigation Framework (Ongoing) [Horizon](#) 

- Researching a hybrid framework combining SLAM-based A* global planning with generative trajectory refinement.
- Implementing a conditional DDPM for collision-anticipating local planning, jointly conditioned on baseline A* waypoints and future environment states predicted by a self-supervised JEPA world model trained on offline data.

Five-Band Audio Equalizer [4 Members]

[Five-Band-Equalizer](#) 

- Designed filters for desired frequency bands in a five-band audio equalizer.
- Developed and simulated the complete analog circuit for the equalizer.
- Tested and debugged the final hardware implementation to ensure desired performance.

Full-Stack E-Commerce Application using MERN

[Ecommerce-App](#) 

- Developing a responsive e-commerce app with MongoDB, Express.js, React, and Node.js.
- Building the frontend using React and Chakra UI for modern, reusable components.
- Implementing backend APIs for authentication, product management, and checkout flow.

Autonomous Mobile Robot (AMR) [10 Members]

[AMR-Platform](#) 

- Built a modular, cost-efficient autonomous mobile robot with LiDAR-based SLAM for path planning for autonomous navigation, closed-loop NEMA 24 steppers, and UDP/Wi-Fi control for indoor navigation.
- Designed the power distribution PCB with optimized grounding, copper pour, and thermal management.
- Performed PCB testing, hardware debugging, and compact wiring integration.

Microphone Preamp with Class AB Power Amplifier [4 Members]

- Built a two-stage audio amplifier with a low-noise microphone preamp and thermally stable Class AB output stage.
- Involved in designing and implementing Class AB amplifier with proper biasing and thermal compensation.

COURSES & CERTIFICATIONS

- **Machine Learning Specialization by Andrew Ng** - Coursera
- **OpenCV Bootcamp** - OpenCV University
- **University Modules** - Pattern Recognition, Image Processing and Machine Vision, Wireless Networks, Electromagnetics, Applies Statics, Linear Algebra, Digital Signal Processing, Signals & Systems

SKILLS AND INTERESTS

- **Programming Languages:** Python, C/C++, JavaScript/TypeScript, MATLAB
- **Frameworks & Libraries:** OpenCV, TensorFlow/Keras, MERN Stack(MongoDB, Express.js, React, Node.js)
- **Software Tools:** Altium Designer, LTSpice, SolidWorks, CubeIDE, Proteus, Arduino, Node-RED
- **Soft Skills:** Problem Solving, Fast Learning, Analytical Skills, Team Work, Work Load Balance
- **Languages:** English: Professional working proficiency, Sinhala: Native Language
- **Interests:** AI/Machine Learning, Computer Vision, Embedded Systems, Telecommunication, Full-Stack Development

COMPETITIONS AND ACHIEVEMENTS

- **First Runners-up** - *Silicon Pulse* Analog Electronics Competition, SLTC 2025
- **Finalist** - IESL Robogames 2025
- **Top 12 Teams** - ElectroCombat Robot Battle 2024
- **Dialog Merit Scholarship** - Awarded for securing 1st rank in Ratnapura District at the GCE Advanced Level Examination 2021(2022)
- **Silver Medalist** - National Mathematics Olympiad 2017

REFERENCES

1. **Prof. Rohan Munasinghe**

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2. **Dr. Sampath K. Perera**

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